

Topic 2: Atoms, elements and compounds

Premium Answer Booklet

53 questions	2020-2025 Paper 4	Premium readable layout
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Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

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Question 01

28 marks

2020 Feb/March Paper 42 Question 3

Main topic: Topic 2: Atoms, elements and compounds
Cross-topic links: Topic 3: Stoichiometry, Topic 8: The Periodic Table
Answer source label: 0620_m20_ms_42.pdf

3(a)	• selenium / Se
3(b)	• Ca has 2 and Cl has outer electrons 7 (1) • Ca (atoms) lose electrons (1) • Cl (atoms) gain electrons (1) • Ca²⁺ (ions) (1) • Cl⁻ (ions) (1)
3(c)(i)	• any number in the range 72 - 129°C
3(c)(ii)	• attraction increase (1) • between molecules (1)
3(c)(iii)	• 3 P - Cl dot cross bonds (1) • 2 (only) non-bonding electrons to make an octet on P (1) • 6 (only) non-bonding electrons to make an octet on each Cl (1)
3(d)(i)	• constant concentrations (1) • rate of forward reaction = rate of reverse reaction (1)
3(d)(ii)	• increased temperature: • (equilibrium) shifts to LHS (1) • (forward) reaction is exothermic (1) • increased pressure: • (equilibrium) shifts to RHS (1) • fewer moles (of gas) on RHS (1)
3(d)(iii)	• rate increases and particles have more energy (1) • more collisions (between particles) occur per second / per unit time • more (of the) particles / collisions have energy greater than activation energy • or • more (of the) particles / collisions have sufficient energy to react • or • a greater percentage / proportion / fraction of collisions (of particles) are successful
3(e)	• mol of LiPF₆ = 3.04 / 152 = 0.02(00) (1) • mol of LiF = 0.02(00) × 6 = 0.12(0) (1) • mass of LiF = 3.12 g (1)
3(f)(i)	• oppositely charged ions • (ions) are attracted
3(f)(ii)	• any two from: • physical constants: high boiling point / melting point • conductivity: conduct (electricity) when aqueous or conduct (electricity) when molten • solubility: soluble in water

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration × volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

Question 02

16 marks

2020 May/June Paper 41 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s20_ms_41.pdf

1(a)(i)	• protons • neutrons
1(a)(ii)	• nucleon number
1(a)(iii)	• No separate point listed.
1(a)(iv)	• 2:8:8
1(a)(v)	• Al_2X_3
1(b)(i)	• isotopes
1(b)(ii)	• ^{12}C
1(b)(iii)	• a mole
1(b)(iv)	• Avogadro constant
1(c)	• $(3 \times 69) + (2 \times 71)$ • 69.8 • = • = • $Y = \text{Ga}$ / gallium
1(d)(i)	• phosphorus / P
1(d)(ii)	• gains electrons • three electrons (when forming ion)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 03

14 marks

2020 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 9: Metals

Answer source label: 0620_s20_ms_41.pdf

2(a)	• metallic (bonding) • sea of electrons • positive ions • attraction between
2(b)(i)	• Mg octet of eight dots • O octet of six crosses and two dots. • correct charges on both ions
2(b)(ii)	• $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ • species • balancing
2(c)(i)	• $2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ • product species • balancing
2(c)(ii)	• (thermal) decomposition
2(c)(iii)	• magnesium carbonate • magnesium hydroxide

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

END OF ANSWER

Question 04

13 marks

2020 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s20_ms_42.pdf

2(a)(i)	• magnesium 2.8 (all crosses) (1) • fluorine 2.8 (seven dots and one cross in outer shell) (1) • Mg^{2+} and F^{-} (1)
2(a)(ii)	• MgF_2
2(a)(iii)	• heat until molten or dissolve in water (1) • moving ions / mobile ions (1)
2(b)	• two single bonds (1) • one double bond (1) • six non-bonding electrons on both F atoms and four non-bonding electrons on O atom to complete the octet in each • case (1)
2(c)(i)	• forces of attraction between oppositely charged ions / ionic bonds (1) • strong / need a lot of energy to break / weaken (1)
2(c)(ii)	• forces of attraction between molecules (1) • weak / need a small of energy to break / weaken (1)

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 05

13 marks

2020 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s20_ms_43.pdf

3(a)(i)	• same number of electrons • same electronic configuration
3(a)(ii)	• number of electrons • number of neutrons • number of protons • $^{35}_{17}\text{Cl}$ • $^{37}_{17}\text{Cl}$ -
3(b)(i)	• displacement / redox
3(b)(ii)	• iodine is less reactive than bromine
3(c)	• magnesium ion has an outer shell with eight crosses • chloride ion has an outer shell with seven dots and one cross • chloride has a charge of 1- and magnesium has a charge 2+
3(d)	• energy needed to break bonds = $436 + 243 = 679$ • energy released when bonds formed = $2 \times 432 = 864$ • energy change = $679 - 864 = -185$

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

END OF ANSWER

Question 06

10 marks

2020 Oct/Nov Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 8: The Periodic Table

Answer source label: 0620_w20_ms_42.pdf

1(a)(i)	• (1) • H (1)
1(a)(ii)	• B
1(a)(iii)	• D
1(a)	• (iv • C and G OR C and E
1(b)	• F (1) • third / outer shell is being filled before second shell is full; second shell has 6 electrons: it should have 8 electrons (1)
1(c)	• No separate point listed.
1(d)(i)	• H-
1(d)(ii)	• aluminium / Al

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 07

7 marks

2020 Oct/Nov Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w20_ms_43.pdf

2(a)	• No separate point listed.
2(b)	• No separate point listed.
2(c)	• 2,8,8
2(d)(i)	• B, C and E
2(d)(ii)	• A
2(d)(iii)	• D
2(d)(iv)	• B and C

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 08

12 marks

2021 Feb/March Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_m21_ms_42.pdf

1(a)(i)	• E
1(a)(ii)	• A • I
1(a)(iii)	• D • G
1(a)(iv)	• F
1(a)(v)	• H
1(v)(i)	• G and I
1(v)(ii)	• A
1(v)(iii)	• B
1(b)	• same proton number • different neutron number

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 09

5 marks

2021 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s21_ms_41.pdf

Answer

- Mg: 12 and 13 (1)
- Cu²⁺: 29 and 27 (1)
- 37(above)
- and 17(below) (1)
- Cl (1)
- 1- (1)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 10

10 marks

2021 May/June Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 8: The Periodic Table, Topic 9: Metals, Topic 7: Acids, bases and salts, Topic 10: Chemistry of the environment

Answer source label: 0620_s21_ms_42.pdf

1(a)	• Na or Mg or Al
1(b)	• Mg
1(c)	• S
1(d)	• S
1(e)	• Al
1(f)	• Ar
1(g)	• Si
1(h)	• Al
1(i)	• Cl
1(j)	• Cl

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 11

19 marks

2021 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s21_ms_42.pdf

2(a)(i)	• isotopes
2(a)(ii)	• mark by row • ^{107}Ag • $^{109}\text{Ag}^+$ • p • n • e
2(a)(iii)	• average / mean • ^{12}C
2(a)(iv)	• 50(%)
2(b)	• HNO_3
2(c)(i)	• yellow precipitate
2(c)(ii)	• $\text{Ag}^+(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{AgI}(\text{s})$ • AgI (as only product) (1) • Ag^+ and I^- (as only reactants) (1) • state symbols (1)
2(d)	• aluminium (1) • ammonia (1)
2(e)	• photochemical
2(f)(i)	• any named alkane
2(f)(ii)	• name of organic product derived from 2(f)(i) (1) • hydrogen chloride / HCl (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO_2 and H_2O when oxygen is in excess.

END OF ANSWER

Question 12

20 marks

2021 May/June Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds
Cross-topic links: Topic 5: Chemical energetics, Topic 7: Acids, bases and salts
Answer source label: 0620_s21_ms_42.pdf

5(a)(i)	• $6\text{Li} + \text{N}_2 \rightarrow 2\text{Li}_3\text{N}$ • N_2 as reactant (1) • rest of equation (1)
5(a)(ii)	• new octet of 8 electrons consisting of 5 crosses and 3 dots in second shell (1) • charge of 3^- (1)
5(b)(i)	• bonds broken $[945 + (3 \times 160)] = 1425$ (1) • bonds formed $(2 \times 3 \times 300) = 1800$ (1) • energy change $= - = 1425 - 1800 = -375$ (1)
5(b)(ii)	• Answer must reflect answer in 5(b)(i) • exothermic • and • more energy released (in bond formation) than used/taken in (in bond breaking)
5(b)(iii)	• N with 1 bonding pair with each F (1) • 2 non-bonding dots for N (1) • 6 non-bonding crosses for F (1)
5(c)	• ionic bonds in Li_3N (1) • attraction between molecules in NF_3 (1) • weaker attraction (between particles) in NF_3 opposite reasoning also allowed (1)
5(d)(i)	• rfm of $\text{NH}_4\text{NO}_3 = 80$ (1) • mass of N $= 2 \times 14 = 28$ and percentage N $= 100 \times 28 / 80 = 35\%$ (1)
5(d)(ii)	• fertiliser
5(d)(iii)	• calcium hydroxide
5(e)(i)	• proton acceptor
5(e)(ii)	• $7 < x \blacksquare 11$

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 13

5 marks

2021 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s21_ms_43.pdf

Answer

- B: 5 and 6 (1)
- Cl -: 18 and 18 (1)
- 54 and 24 (1)
- Cr (1)
- 3+ (1)
- Question Answer

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 14

22 marks

2021 Oct/Nov Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 3: Stoichiometry

Answer source label: 0620_w21_ms_41.pdf

2(a)(i)	• isotopes
2(a)(ii)	• 1 mark for each correct row • ^{63}Cu • $^{65}\text{Cu}^{2+}$ • p • n • e
2(a)(iii)	• $= (70 \times 63) + (30 \times 65)$ or $[(4410) + (1950)]$ or 6360 (1) • $= / 100 = 63.6$ (1) • OR • $= (0.7(0) \times 63) + (0.3(0) \times 65)$ or $[(44.1(0)) + (19.5(0))]$ (1) • $= 63.6$ (1)
2(b)(i)	• white (1) • to (light) blue (1)
2(b)(ii)	• $5\text{H}_2\text{O}$
2(b)(iii)	• heating
2(b)(iv)	• boiling point (1) • is $100\text{ }^{\circ}\text{C}$ • OR • freezing point (1) • is $0\text{ }^{\circ}\text{C}$ (1)
2(c)(i)	• blue precipitate
2(c)(ii)	• Alternative suggestion: • $\text{Cu}(\text{OH})_2$ (as only product) (1) • Cu^{2+} and 2OH^- (as reactants) (1) • state symbols (1)
2(d)	• 188 • $4.7 / 188 = 0.025(0)$ • $0.025(0) \times 2 = 0.05(0)$ • $0.05(0) \times 24.0 = 1.2$
2(e)	• $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$ • NaNO_2 (1) • rest of equation (1)

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates
- clearly stated.

END OF ANSWER

Question 15

8 marks

2021 Oct/Nov Paper 42 Question 3

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w21_ms_42.pdf

3(a)

- 1 mark for each correct row
- Name
- Relative mass
- Relative charge
- Proton
- +1
- Neutron
- Electron
- 1 / 1840
- -1

3(b)

- Particle
- Number of
- protons
- Number of
- neutrons
- Number of
- electrons
- $^{32}_{16}\text{S}$
- $^{39}_{19}\text{K}^{+}$
- $^{79}_{35}\text{Br}^{-}$
- = row 1 (1) = row 2 (1) = Br (1)
- = 7935 (on left of any symbol) (1) = charge (on any symbol) (1)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 16

6 marks

2021 Oct/Nov Paper 43 Question 4

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w21_ms_43.pdf

4(a)	• 1 mark for each row
4(b)	• diamond and graphite
4(c)(i)	• $(22 / 44 =) 0.5$ (moles) (1) • 3.01×10^{23} (1)
4(c)(ii)	• 1.505×10^{23}

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 17

10 marks

2022 Feb/March Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_m22_ms_42.pdf

1(a)	• nitrogen
1(b)	• sodium
1(c)	• iron
1(d)	• silicon
1(e)	• chlorine
1(f)	• cobalt
1(g)	• chlorine
1(h)	• copper
1(i)	• oxygen
1(j)	• zinc

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 18

8 marks

2022 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s22_ms_41.pdf

2(a)(i)	• charge • relative mass • -1 • (1) • (1) • Mark by column
2(a)(ii)	• 12 (1) • 12 (1) • Mark by row
2(a)(iii)	• (they have) 2 more protons than electrons • OR • (they have) 2 fewer electrons than protons • OR • (they have) 12 protons and 10 electrons
2(b)	• Na⁺ or Al³⁺ (1) • F⁻ or O²⁻ or N³⁻ (1) • Ne (1)

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 19

14 marks

2022 May/June Paper 42 Question 3

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 8: The Periodic Table

Answer source label: 0620_s22_ms_42.pdf

3(a)(i)	• atom(s) of the same element • with different number of neutrons
3(a)(ii)	• No separate point listed.
3(a)(iii)	• 24 protons; 28 neutrons; 21 electrons
3(b)(i)	• blue • pink
3(b)(ii)	• (act as) catalysts • variable oxidation state
3(c)(i)	• mobile electrons
3(c)(ii)	• malleability / malleable
3(d)	• high(er) density • high(er) melting points

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 20

13 marks

2022 May/June Paper 42 Question 4

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 3: Stoichiometry

Answer source label: 0620_s22_ms_42.pdf

4(a)	• (pale) yellow and gas
4(b)	• Method 1 • S 32.1 / 32 or 0.78/0.79 ... • and • F 19.0 / 19 or 3.93 / 3.94... • / both by 0.7875 = 1 : 5 or SF₅ • (32 / 19 = 2 and) S₂F₁₀ • Method 2 • 32 -> 32.1/100 and 19 -> 19.0 / 100 OR '64' and '190' • 64 / 32 and 190 / 19 • (2 and 10) to give S₂F₁₀
4(c)	• N with 1 dot-cross bonding pair with each Cl • 2 non-bonding dots for N • 6 non-bonding crosses for Cl
4(d)	• two crosses in first shell of Li • 7 dots and 1 cross in third shell of Cl • '+' charge on Li on correct answer line and '-' charge on Cl ion on correct answer line
4(e)	• ionic bonds in LiCl • attraction between molecules in NCl₃ • weaker attraction (between particles) in NCl₃ opposite reasoning also allowed

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 21

8 marks

2022 May/June Paper 43 Question 1

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 9: Metals

Answer source label: 0620_s22_ms_43.pdf

1(a)	• carbon dioxide
1(b)	• iron(III) oxide
1(c)	• copper
1(d)	• carbon monoxide
1(e)	• glucose
1(f)	• carbon monoxide
1(g)	• copper
1(h)	• oxygen

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.
- Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates
- clearly stated.

END OF ANSWER

Question 22

8 marks

2022 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s22_ms_43.pdf

2(a)(i)	• number of protons (are the same) / 16 protons (1) • number of electrons (are the same) / 16 electrons (1) • number of neutrons (are different) / 16,17 neutrons (1)
2(a)(ii)	• number of protons is the same as (the number of) electrons
2(a)(iii)	• same number of (outer shell) electrons
2(b)(i)	• (they have) two more electrons than protons
2(b)(ii)	• P³⁻ OR Cl⁻ (1) • K⁺ OR Ca²⁺ (1)

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 23

12 marks

2022 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 3: Stoichiometry

Answer source label: 0620_s22_ms_43.pdf

3(a)(i)	• covalent
3(a)(ii)	• weak force(s) of attraction between molecules
3(a)(iii)	• no ions • OR no mobile electrons
3(b)	• 450 °C (1) • 200 atmospheres (1) • iron (catalyst) (1) • $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (1)
3(c)	• $2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \rightarrow 2\text{NH}_3 + \text{CaCl}_2 + 2\text{H}_2\text{O}$
3(d)	• all 4 NH dot and cross bonds (1) • single bonding pair between N's and two non-bonding electrons on each N and no non-bonding e on H and nitrogen octet • complete (1)
3(e)(i)	• proton acceptor
3(e)(ii)	• $\text{N}_2\text{H}_4 + \text{H}_2\text{O} \rightleftharpoons \text{N}_2\text{H}_5^+ + \text{OH}^-$

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 24

8 marks

2022 Oct/Nov Paper 41 Question 1

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 10: Chemistry of the environment, Topic 8: The Periodic Table, Topic 7: Acids, bases and salts

Answer source label: 0620_w22_ms_41.pdf

1(a)	• oxygen
1(b)	• carbon
1(c)	• lithium
1(d)	• nitrogen
1(e)	• neon • 1f • lithium
1(g)	• boron
1(h)	• nitrogen

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 25

8 marks

2022 Oct/Nov Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w22_ms_42.pdf

1(a)	• No separate point listed.
1(b)	• giant covalent
1(c)	• silicon dioxide
1(d)	• layers • hexagon(al) (rings of carbon)
1(e)	• mobile electrons
1(f)	• No separate point listed.
1(g)	• limewater

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates
- clearly stated.

END OF ANSWER

Question 26

8 marks

2022 Oct/Nov Paper 43 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w22_ms_43.pdf

1(a)

- relative
- charge
- relative
- mass
- +1
- 1 mark for each correct column

1(b)

- 22 (1)
- 17 (1)
- 17 (1)
- 32 and 16 (1)
- S (1)
- 2 - / - 2 / - - (1)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 27

16 marks

2023 Feb/March Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 7: Acids, bases and salts

Answer source label: 0620_m23_ms_42.pdf

2(a)	• metallic
2(b)(i)	• lighted splint and (squeaky) pop
2(b)(ii)	• No separate point listed.
2(b)(iii)	• universal indicator • February/March 2023
2(b)(iv)	• $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$ • NaOH as product in equation (1) • fully correct equation (1) • state symbols (1)
2(c)(i)	• isotope(s)
2(c)(ii)	• 6Li • 7Li^+ • protons • neutrons • electrons • each row ✓
2(c)(iii)	• $(6 \rightarrow 10) + (7 \rightarrow 90) (= 690)$ (1) • $690 / 100 = 6.9$ (1)
2(d)	• eight dots in third shell of both K (1) • six crosses and two dots in second shell of O (1) • '+' charge on each K on correct answer line and '2-' charge on O ion on correct answer line (1) • February/March 2023

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 28

10 marks

2023 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s23_ms_41.pdf

2(a)(i)	• N
2(a)(ii)	• B
2(a)(iii)	• F
2(a)(iv)	• C
2(a)(v)	• Li
2(a)(vi)	• Ne
2(b)(i)	• different atoms of the same element with the same number of protons(1) • different numbers of neutrons(1)
2(b)(ii)	• $10 \rightarrow 20 +$ • $11 \rightarrow 80$ • $(= 1080)(1)$ • $(1080 / 100 =) 10.8(1)$

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 29

9 marks

2023 May/June Paper 41 Question 3

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s23_ms_41.pdf

3(a)(i)	• Na with 2,8 all crosses(1) • O with 2,8 outer shell with 6 dots and 2 crosses(1) • + AND 2■(1)
3(a)(ii)	• Na₂O
3(b)	• both bonds with 2 dots and 2 crosses(1) • 2 lone pairs • (all dots or all crosses) on both oxygen atoms completing all 3 octets(1)
3(c)(i)	• positive ions and negative ions (1) • strong attraction / strong bonds (1)
3(c)(ii)	• intermolecular forces

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.

END OF ANSWER

Question 30

18 marks

2023 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 8: The Periodic Table

Answer source label: 0620_s23_ms_42.pdf

2(a)	• halogen(s)
2(b)	• same number of outer shell electrons
2(c)	• gas • pale yellow-green • gas • red-brown • both gases
2(d)(i)	• nucleon number / mass number
2(d)(ii)	• ^{79}Br • ^{81}Br- • protons • neutrons • electrons • Each row ✓
2(d)(iii)	• $79 \rightarrow 55(\%) + 81 \rightarrow 45(\%)$ • $7990 / 100 = 79.9$
2(e)(i)	• $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$ • KCl as product • correct equation
2(e)(ii)	• chlorine less reactive than fluorine
2(f)(i)	• white precipitate
2(f)(ii)	• $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$ • AgCl as only product • $\text{Ag}^+ + \text{Cl}^-$ as only reactants (in 1 : 1 ratio) • state symbols

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.

END OF ANSWER

Question 31

11 marks

2023 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s23_ms_43.pdf

2(a)(i)	• Al
2(a)(ii)	• Ar
2(a)(iii)	• Cl
2(a)(iv)	• Al
2(a)(v)	• S
2(a)(vi)	• Cl
2(b)(i)	• 12C
2(b)(ii)	• 24 -> 85(%) + 26 -> 15(%) • $2430 / 100 = 24.3$
2(c)(i)	• No separate point listed.
2(c)(ii)	• Aluminium / Al

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 32

10 marks

2023 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s23_ms_43.pdf

3(a)(i)	• eight crosses in second shell of Mg • 7 dots and 1 cross in second shell of F • '2+' charge on Mg ion on correct answer line • and '-' charge on F ion on correct answer line
3(a)(ii)	• MgF₂
3(a)(iii)	• melting
3(b)	• 4 dot and cross single bonds • 3 pairs of non-bonding e on each Cl and no non-bonding e on Si
3(c)(i)	• molecule(s)
3(c)(ii)	• covalent bonds • strong bonds • and • giant (covalent) structure

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.

END OF ANSWER

Question 33

21 marks

2023 Oct/Nov Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 4: Electrochemistry

Answer source label: 0620_w23_ms_41.pdf

2(a)	• 5p and 5e (1) • 6n (1)
2(b)(i)	• 20%
2(b)(ii)	• $0.540 / 10.8 = 0.05(0)$ mol (1) • $0.05 \rightarrow 6.02 \rightarrow 1023 = 3.01 \rightarrow 1022$ (1)
2(c)(i)	• bauxite
2(c)(ii)	• cryolite (1) • lowers operating temperature OR improves conductivity (1)
2(c)(iii)	• $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ • $\text{Al}^{3+} + 3\text{e}^-$ on the left-hand side (1) • equation correct (1)
2(c)(iv)	• (anodes of) carbon react with oxygen (formed at the anode) (1) • (form) carbon dioxide (1)
2(d)	• (good) conductor (of electricity) (1) • low density (1)
2(e)	• aluminium oxide layer (1) • (oxide layer) is unreactive (1)
2(f)(i)	• $2\text{Al} + 3\text{F}_2 \rightarrow 2\text{AlF}_3$ • Al and F₂ as reactants (1) • equation correct (1)
2(f)(ii)	• eight crosses in second shell of Al (1) • 7 dots and 1 cross in second shell of F (1) • 3+ charge on Al ion on correct answer line and - charge on F ion on correct answer line (1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 34

8 marks

2023 Oct/Nov Paper 42 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w23_ms_42.pdf

1(a)	• B
1(b)	• F
1(c)	• D
1(d)	• AND E
1(e)	• G
1(f)	• C
1(g)	• B
1(h)	• F

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 35

14 marks

2023 Oct/Nov Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w23_ms_42.pdf

2(a)(i)	• ^{59}Co • $^{65}\text{Cu}^{2+}$ • protons • neutrons • electrons • one mark for each correct row
2(a)(ii)	• $(63 \rightarrow 70) + (65 \rightarrow 30) = 6360$ (1) • $6360 / 100 = 63.6$ (1)
2(b)	• (high) density (1) • (high) melting point (1)
2(c)(i)	• water molecules (1) • (the water present) in hydrated crystals (1)
2(c)(ii)	• pink (1) • $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (1)
2(c)(iii)	• white (1) • blue (1)
2(c)(iv)	• heating (the hydrated copper(II) sulfate)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates
- clearly stated.

END OF ANSWER

Question 36

8 marks

2023 Oct/Nov Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w23_ms_43.pdf

2(a)(i)	• A, B, F
2(a)(ii)	• D
2(a)(iii)	• C
2(b)	• No separate point listed.
2(c)	• No separate point listed.
2(d)	• 2,8,8
2(e)	• No separate point listed.
2(f)	• No separate point listed.

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 37

5 marks

2024 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s24_ms_41.pdf

Answer

- : Cl : 17 AND 20 (1)
- Cu+: 29 AND 28 (1)
- 33(above) and 16(below) on left hand side of symbol (1)
- S (1)
- 2- (1)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 38

14 marks

2024 May/June Paper 41 Question 3

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 4: Electrochemistry

Answer source label: 0620_s24_ms_41.pdf

3(a)	• $2\text{Na} + \text{F}_2 \rightarrow 2\text{NaF}$(2) • NaF(1) • Has to be the only product • equation completely correct(1)
3(b)(i)	• electrons move • OR • electrons are mobile • OR • electrons flow
3(b)(ii)	• one shared dot and cross(1) • 6 non-bonding electrons (either) dots or crosses on each fluorine atom to complete both octets (1)
3(b)(iii)	• liquid(1) • BOTH melting point is below $-200\text{ }^{\circ}\text{C}$ AND boiling point is above $-200\text{ }^{\circ}\text{C}$(1) • OR • BOTH $-200\text{ }^{\circ}\text{C}$ is higher than $-220\text{ }^{\circ}\text{C}$/ melting point AND lower than $-188\text{ }^{\circ}\text{C}$/ boiling point(1) • OR • $-200\text{ }^{\circ}\text{C}$ is between melting point or $-220\text{ }^{\circ}\text{C}$ and boiling point or $-188\text{ }^{\circ}\text{C}$(1)
3(b)(iv)	• ionic bonds in NaF(1) • attraction between molecules or intermolecular forces in F_2(1) • weaker attraction (between particles) in F_2 opposite reasoning also allowed (1)
3(c)(i)	• breakdown by (the passage of) electricity(1) • of an ionic compound in molten or aqueous (state) (1)
3(c)(ii)	• $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ • $\text{H}^+ + \text{e}^-$ on left hand side(1) • equation fully correct(1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 39

17 marks

2024 May/June Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s24_ms_42.pdf

5(a)(i)	• 32S • 34S • protons • neutrons
5(a)(ii)	• same electronic configuration / structure
5(a)(iii)	• 34 and g
5(a)(iv)	• 1 mole
5(a)(v)	• $(32 \rightarrow 95) + (34 \rightarrow 5) = 3210$ • $3210 / 100 = 32.1$
5(b)(i)	• eight crosses in second shell of Mg • 6 dots and 2 cross in third shell of S • '2+' charge on Mg on correct answer line • and '2-' charge on S on correct answer line
5(b)(ii)	• ionic bonds are strong
5(b)(iii)	• mobile ions
5(c)(i)	• H₂SO₃
5(c)(ii)	• SO₃²⁻
5(d)(i)	• the oxidation number of manganese / Mn is +7
5(d)(ii)	• purple • to • colourless

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 40

5 marks

2024 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s24_ms_43.pdf

Answer

- Cu: 29 and 34
- Cl⁻ : 17 and 18
- 6430 to left of symbol
- Zn
- 2+

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 41

14 marks

2024 May/June Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 4: Electrochemistry

Answer source label: 0620_s24_ms_43.pdf

3(a)(i)	mobile electrons
3(a)(ii)	- double bond of two dots and two crosses 4 non-bonding electrons on each oxygen atom
3(a)(iii)	- liquid - melting point is below $-195\text{ }^{\circ}\text{C}$ and boiling point is above $-195\text{ }^{\circ}\text{C}$ - or $-195\text{ }^{\circ}\text{C}$ is higher than $-199\text{ }^{\circ}\text{C}$ / melting point and lower than $-191\text{ }^{\circ}\text{C}$ / boiling point - or $-195\text{ }^{\circ}\text{C}$ is in between melting point / $-199\text{ }^{\circ}\text{C}$ and boiling point / $-191\text{ }^{\circ}\text{C}$
3(a)(iv)	- graphite has a giant covalent structure - attraction between molecules in carbon monoxide - weaker attraction (between particles) in carbon monoxide opposite reasoning also allowed
3(b)	$2\text{K} + \text{Cl}_2 \rightarrow 2\text{KCl}$ - KCl as only product - correct equation
3(c)(i)	- breakdown by (the passage of) electricity - of an ionic compound in molten or aqueous (state)
3(c)(ii)	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ - any negatively charged OH species losing electrons - correct ionic half-equation

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 42

14 marks

2024 Oct/Nov Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w24_ms_41.pdf

2(a)	total number of protons and neutrons in the nucleus of an atom
2(b)	they are the same
2(c)	2,8,4 2,8,8
2(d)	$12n + 11e$ 10e - Ga 69 above 31 and to the left of the symbol - charge of 3 +
2(e)(i)	- atom(s) of the same element - with different number of neutrons
2(e)(ii)	(3 → 203) AND (7 → 205) (= 2044) 204.4
2(e)(iii)	same number of electrons / same electronic configuration

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 43

16 marks

2024 Oct/Nov Paper 43 Question 3

Main topic: Topic 2: Atoms, elements and compounds
Cross-topic links: Topic 7: Acids, bases and salts, Topic 3: Stoichiometry
Answer source label: 0620_w24_ms_43.pdf

3(a)	• 78.8 (%)
3(b)(i)	• reacts with acids and with bases to produce a salt and water
3(b)(ii)	• $\text{SnO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SnO}_3 + \text{H}_2\text{O}$ (2) • Na_2SnO_3 • correct equation
3(c)	• metals form ionic compounds or ionic bonds only • OR • covalent compounds contain non-metals only
3(d)(i)	• four single bonds using one dot and one cross • three pairs of non-bonding electrons on each chlorine and no non-bonding electrons on tin
3(d)(ii)	• attraction between molecules or intermolecular forces in tin(IV) chloride • tin(II) oxide has a giant ionic structure • weaker attraction (between particles) in tin(IV) chloride opposite reasoning also allowed
3(e)(i)	• unreactive coating of aluminium oxide
3(e)(ii)	• no reaction • $\text{Mg} + \text{Sn}^{2+} \rightarrow \text{Mg}^{2+} + \text{Sn}$
3(f)(i)	• substance that is chemically combined with water • OR • containing water of crystallisation
3(f)(ii)	• $2\text{Sn}(\text{NO}_3)_2 \cdot 20\text{H}_2\text{O} \rightarrow 2\text{SnO} + 4\text{NO}_2 + \text{O}_2 + 40\text{H}_2\text{O}$ (2) • $2(\text{SnO}) + 4(\text{NO}_2)$ • $40\text{H}_2\text{O}$

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 44

16 marks

2025 Feb/March Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 4: Electrochemistry

Answer source label: 0620_m25_ms_42.pdf

2(a)	• (strong) (electrostatic) attraction between ions • ions have opposite charges
2(b)	• 8 dots in third shell of each K • 6 crosses and 2 dots in third shell of S • '+' charge on each K ion on correct answer line • and '2-' charge on S ion on correct answer line
2(c)(i)	• 3 correctly positioned positive ions next to given negative • all ions correct
2(c)(ii)	• cation(s)
2(d)(i)	• electrolysis
2(d)(ii)	• $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$ • any negative Br species losing electron(s) • correct ionic half equation • February/March 2025
2(d)(iii)	• hydrogen • bubbles • oxygen water • bubbles

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 45

17 marks

2025 Feb/March Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 3: Stoichiometry

Answer source label: 0620_m25_ms_42.pdf

5(a)	• 55Mn • 42Ca^{2+} • protons • neutrons • electrons • Each row ✓
5(b)(i)	• (Mn is a) transition element
5(b)(ii)	• variable oxidation state
5(b)(iii)	• MnO
5(c)(i)	• a substance that reduces another substance and is itself oxidised • 'a substance' reduces another substance • is itself oxidised
5(c)(ii)	• $3\text{Mn}_3\text{O}_4 + 8\text{Al} \rightarrow 4\text{Al}_2\text{O}_3 + 9\text{Mn}$ • Al_2O_3 • correct equation
5(d)(i)	• $50 \rightarrow 0.200/1000 = 0.01(00)$ • $0.01/4 = 0.0025(0) / 2.5(0) \rightarrow 10^{-3}$ • $= 0.0025 \rightarrow 24000 = 60(0)$
5(d)(ii)	• (damp) litmus (paper) • and • is bleached / goes white • February/March 2025
5(d)(iii)	• kinetic energy of particles decreases • frequency of collisions between particles decreases • lower percentage / proportion / fraction of collisions / particles have energy greater than / equal to activation energy • OR • Fewer of the collisions / particles have energy greater than / equal to activation energy

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 46

10 marks

2025 May/June Paper 41 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s25_ms_41.pdf

2(a)	• charge • (1) • mass(1) • -1 • +1 • 1 mark for each correct column
2(b)(i)	• electron • neutron • proton • 1 mark for each correct column
2(b)(ii)	• $(41 \times 10) + (39 \times 90) = 3920$ (1) • $(3920 / 100) = 39.2$(1)
2(b)(iii)	• aluminium has only 1 isotope which has nucleon number 27
2(c)	• Ar (1) • P3- or S2- or Cl - (1)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 47

16 marks

2025 May/June Paper 42 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s25_ms_42.pdf

2(a)	• pair of electrons • electron(s) shared between two atoms
2(b)(i)	• the oxidation number of chlorine / Cl • (oxidation number) is +1
2(b)(ii)	• O with 1 dot-cross bonding pair with each Cl • 4 non-bonding dots for O • 6 non-bonding crosses for each Cl
2(b)(iii)	• (forces of) attraction between molecules are broken (during boiling) • covalent bonds (between Cl and O atoms) are broken (during thermal decomposition) • weak attraction between molecules and strong covalent bonds / bonding between atoms correctly applied
2(b)(iv)	• no ions • no mobile electrons
2(c)(i)	• graphite
2(c)(ii)	• electrons
2(c)(iii)	• 5 correct Si atoms • 4 correct O atoms

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 48

14 marks

2025 May/June Paper 42 Question 5

Main topic: Topic 2: Atoms, elements and compounds

Cross-topic links: Topic 9: Metals

Answer source label: 0620_s25_ms_42.pdf

5(a)	• I
5(b)(i)	• lithium / Li
5(b)(ii)	• francium / Fr
5(b)(iii)	• lithium / Li
5(b)(iv)	• potassium / K
5(b)(v)	• potassium / K
5(c)	• 8 dots in second shell of each Na • 6 crosses and 2 dots in third shell of S • '+' charge on each Na ion on correct answer line and '2-' charge on S ion on correct answer line
5(d)(i)	• isotopes
5(d)(ii)	• mark by row • p = 37 and 37 • n = 48 and 50 • e = 37 and 36
5(d)(iii)	• 75(%)

Easy explanation / reminder

• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 49

10 marks

2025 May/June Paper 43 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Cross-topic links: Topic 11: Organic chemistry, Topic 9: Metals
Answer source label: 0620_s25_ms_43.pdf

1(a)	• silicon(IV) oxide
1(b)	• propene
1(c)	• aluminium oxide
1(d)	• graphite
1(e)	• propene • OR • ethanol
1(f)	• silicon(IV) oxide • calcium oxide
1(g)	• propane
1(h)	• ethanol
1(i)	• nitrogen

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 50

9 marks

2025 May/June Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_s25_ms_43.pdf

2(a)	• protons and neutrons
2(b)(i)	• electron • neutron • proton • 1 mark for each correct column
2(b)(ii)	• $(32 \rightarrow 95) + (34 \rightarrow 5) = 3210$ • $3210 / 100 = 32.1$
2(b)(iii)	• ^{12}C
2(c)	• Ne • Na^+ OR Mg^{2+} OR Al^{3+}

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 51

13 marks

2025 Oct/Nov Paper 41 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w25_ms_41.pdf

1(a)	• particle • relative • mass • relative • charge • proton • neutron • 0 / nil • electron • 1 • ✓(1) • ✓(1)
1(b)(i)	• atoms of the same element that have the same number of protons • but different numbers of neutrons
1(b)(ii)	• they have the same number of electrons and therefore the same electronic configuration
1(c)	• 22 • 16 and 18 • 50 and 22 and to the left of the symbol • Ti • 2+
1(d)	• nucleon number
1(e)	• mol Ar = $2.0 \div 40 = 0.05$ (mol) • number of atoms = $0.05 \rightarrow 6.02 \rightarrow 10^{23} = 3.01 \rightarrow 10^{22}$

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 52

9 marks

2025 Oct/Nov Paper 43 Question 1

Main topic: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w25_ms_43.pdf

1(a)	• neutron • proton • +1
1(b)	• 18 • 10 • 17 above 8 to the left of the symbol • O • 2- / -2 / - -
1(c)(i)	• 12C / carbon-12
1(c)(ii)	• 85 (%)

Easy explanation / reminder

- Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 53

8 marks

2025 Oct/Nov Paper 43 Question 2

Main topic: Topic 2: Atoms, elements and compounds
Cross-topic links: Topic 1: States of matter
Answer source label: 0620_w25_ms_43.pdf

2(a)(i)	• B
2(a)(ii)	• F
2(a)(iii)	• D
2(b)(i)	• A • high melting point AND/OR high boiling point • poor conductor of electricity when solid AND molten
2(b)(ii)	• C • only conducts electricity when molten

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER